MATE 664 L09
Atomic Model For Diffusion (II)

Recap: we have seen link between atomistic & macropic diffusivities!

Einstein equation

$$\langle R^2 \rangle = \frac{\int_0^\infty r^2 c(r,t)\, 4\pi r^2 dr}{\int_0^\infty c(r,t)\, 4\pi r^2 dr} = \begin{matrix} 6Dt & (3D) \\ 4Dt & (2D) \\ 2Dt & (1D) \end{matrix}$$

From random walk picture

1D

$$\langle R^2 \rangle = N_t \langle r^2 \rangle = \Gamma \tau \langle r^2 \rangle$$

τ: time; Γ: Frequency; $\langle r^2 \rangle$: mean jump dist^2

1D: $\langle R^2 \rangle = \Gamma \tau \langle r^2 \rangle = 2Dt$ (same)

$$\boxed{D = \frac{\Gamma \langle r^2 \rangle}{2}}$$

The equation $D = \frac{\Gamma \langle r^2 \rangle}{2}$ from random walk is actually striking!

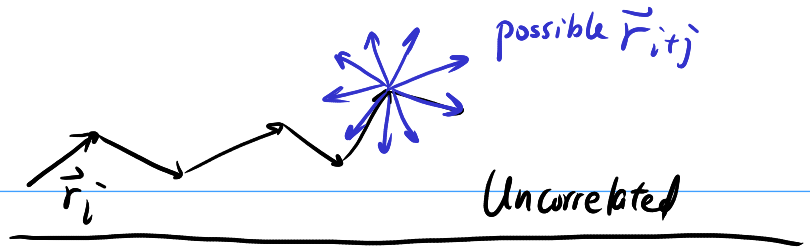
D: unit m^2/s
Γ: frequency $1/s$
$\langle r^2 \rangle$: mean sq dist m^2

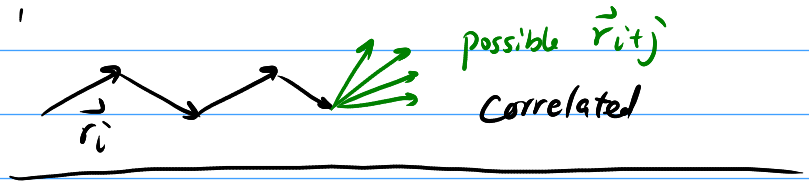
$\Rightarrow$ direct linking unit meaning

The random walk result is a consequence of "unrelated" jumps, for "correlated" jumps we need a factor

$$\langle R^2 \rangle = N_\tau \langle r^2 \rangle f$$

$$f = 1 + \frac{2}{N_\tau(r)} \left\langle \sum_{j=1}^{N_\tau - 1} \sum_{i=1}^{N_\tau - j} |\vec{r}_i| |\vec{r}_{i+j}| \cos\theta_{i,i+j} \right\rangle$$





General term $D = \frac{\Gamma \langle r^2 \rangle}{6} f$ (3D)

The random walk picture can be applied for

- gas !
- liquid !
- solid !

Gas molecules: kinetic theory of gas

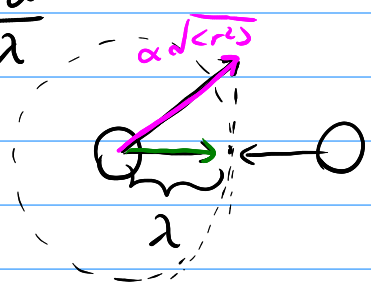
For gas molecules A & A^* (labelled, same species)

average thermal velocity $\bar{u} = \sqrt{\frac{8 k_B T}{\pi m}}$

mean free path $\lambda = \frac{1}{\sqrt{2}\pi d^2 n}$

Collision frequency $\Gamma = \frac{\bar{u}}{\lambda}$

Also $\langle r^2 \rangle = 2\lambda^2$



$$\Rightarrow D = \frac{\Gamma \langle r^2 \rangle}{6}$$

$$= \frac{\frac{\bar{u}}{\lambda} \cdot 2\lambda^2}{6} = \frac{1}{3}\bar{u}\lambda$$

n: number density

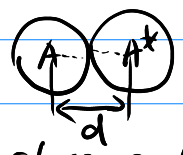
$p = nRT \Rightarrow \frac{p}{RT} = n$ $\quad D \propto \sqrt{T} \cdot \frac{1}{p/RT} = T^{1/2}/p$

The power law $\begin{cases} D_{AA^*} \propto T^{1.5}/p & \text{self-diffusivity} \\ D_{AB} \propto T^{1.5}/p & \text{intrinsic diffusivity} \end{cases}$

correspond to the Chapman-Enskog theory.

Question: why haven't we see the $D \propto \exp(-\frac{E}{k_B T})$ term?

The distance ($\langle x^2 \rangle$ or λ^2 depends on density)



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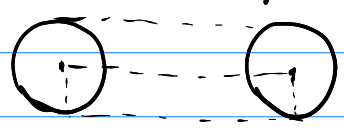
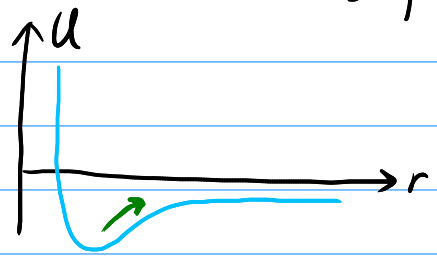


$$\text{density} = \frac{N^\#}{\text{Volume}}$$

Volume ≈ Cross-section × length

$$\lambda \propto \frac{1}{\pi d^2 \cdot n}$$ with some factor

$\begin{cases} \text{Thermal activation} \Rightarrow \text{faster } \bar{u} \Rightarrow \text{higher } \Gamma \checkmark \\ \text{Energy landscape? Lennard-Jones potential} \end{cases}$

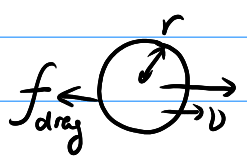


No barrier to overcome!

Diffusion in Liquid

We still have $D = \frac{\Gamma \langle r^2 \rangle}{6}$

Einstein equation for particle



Previous lecture 4

$D = \mu k_B T$

↑ mobility

How do we get M ? Fluid use Stokes law

$$M = \frac{[\text{velocity}]}{[\text{Driving Force}]} = \frac{[\text{velocity}]}{[\text{Drag Force}]} = \frac{v}{f_{drag}}$$

$$f_{drag} = 6\pi \eta r \cdot v$$

viscosity

$\Rightarrow$ Stokes-Einstein equation

$$D = M k_B T = \frac{v}{f_{drag}} \cdot k_B T = \frac{1}{6\pi \eta r} \cdot k_B T$$

size of particle

What about the $\langle R^2 \rangle$ picture ? (KOM chapter 10.5.2)

Newton's eqn for particle mass m, friction factor f

$$m \cdot a = F_x - f v$$

D.F. in x

$$m \frac{d^2x}{dt^2} = F_x - f \frac{dx}{dt}$$

we're interested in $\langle x^2 \rangle$

$$\frac{d(x^2)}{dt} = 2x \frac{dx}{dt}$$

$$\frac{d}{dt}\left(\frac{d(x^2)}{dt}\right) = \frac{d\left(2x \frac{dx}{dt}\right)}{dt} = 2x \frac{d^2x}{dt^2} + 2\left(\frac{dx}{dt}\right)^2$$

Final exp:

$$\frac{m}{2} \frac{d}{dt}\left[\frac{d(x^2)}{dt}\right] - m\left(\frac{dx}{dt}\right)^2 = -\frac{f}{2}\frac{d(x^2)}{dt} + x F_x$$

Expected value

$$\underbrace{\frac{m}{2}\left\langle \frac{d}{dt}\left[\frac{d(x^2)}{dt}\right]\right\rangle}_{\frac{d}{dt}\left\langle \frac{d(x^2)}{dt}\right\rangle} - \underbrace{m\left\langle\left(\frac{dx}{dt}\right)^2\right\rangle}_{kT} = -\frac{f}{2}\left\langle \frac{d(x^2)}{dt}\right\rangle + \underbrace{\langle x F_x \rangle}_{0}$$

avg $x \to 0$

Simplify $\Rightarrow$ $\frac{m}{2}\frac{d}{dt}\left\langle\frac{d(x^2)}{dt}\right\rangle + \frac{f}{2}\left\langle\frac{d(x^2)}{dt}\right\rangle = kT$

$$\Rightarrow \left\langle\frac{d(x^2)}{dt}\right\rangle = \underbrace{Ae^{-\frac{ft}{m}}}_{\text{decays, long time} \Rightarrow 0} + \frac{2kT}{f} \quad \text{general solution}$$

$$\therefore \left\langle\frac{d(x^2)}{dt}\right\rangle = \frac{2kT}{f} \Rightarrow \langle x^2\rangle = \frac{2kT}{f}\cdot t$$

$$\langle R^2\rangle = \langle x^2\rangle + \langle y^2\rangle + \langle z^2\rangle = \frac{6kT}{f}\cdot t$$

Use Einstein relation of D

$$D = \frac{\Gamma\cdot\langle R^2\rangle}{6} = \frac{\cancel{\Gamma}\cdot\frac{\cancel{6}kT}{f}\cdot\cancel{t}}{\cancel{6}} = \frac{kT}{f}$$

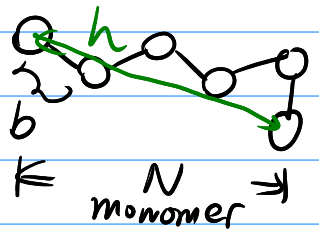
In other words, Einstein relation in mean medium with friction

$$D = \frac{k_BT}{f} = \mu\cdot k_BT \qquad \text{mobility} = \frac{1}{\text{friction factor}}$$

$$f_{drag} = f\cdot v \qquad f = \frac{1}{\mu}\,!$$

$$F\mu = v$$

Liquid / Polymer diffusivity can move from same picture, with modifications



End to end random walk (no angle constraint)

$\langle h^2\rangle = Nb^2 \Rightarrow$ polymer shape is not linear !

Effective radius

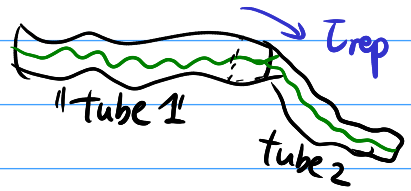
$R_h \propto \sqrt{\langle h^2\rangle}$ $\nearrow N^{3/5}b$ (good solvent)

$\searrow N^{1/2}b$ (theta solvent)

For polymer chain with R_h Stokes–Einstein Eq

self diffusivity $\rightarrow$ $D^* \propto \frac{kT}{6\pi\eta R_h}$ (may have correction factor)

For densely packed polymer chain, D^* can explain the time scale for process like reptition



$$^*D = \frac{\nu_0}{6}\left(\frac{u}{N}\right)^2$$

u: each excursion step

N: polymer units

polymer with isolated chain $\propto \frac{1}{N^2}$

Stokes–Einstein eq means $D \propto \frac{1}{\eta}$ diffusivity & viscosity inverse!

Empirically $\mu = Ae^{B/T}$ (BSL book)

we should also expect $D \propto T \cdot e^{-B/T}$ also motivated

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Diffusion model in solids

- Multiple mechanisms exist (What is mechanism? abstracted atomic motion)

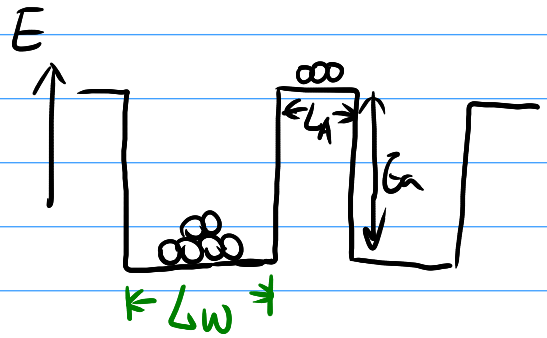
- Mechanism depends on lattice / bonding / size difference / charge

- Species to diffuse?
  1. substitutional "replace sites"
  2. interstitial "between sites"

Substitutional: vacancy take over
Interstitial: solute atom squeeze in
Both encounters a "Barrier"!

Remember our goal is still $D = \frac{\Gamma \langle r^2 \rangle}{6}$
there are simplified models to choose from

Generalized well potential (where does well come from?)



From Boltzmann distribution

$$\frac{P_{activated}}{P_{well}} = \exp\left(-\frac{E_a}{k_B T}\right)$$

1) Γ: how frequent do one particle move over barrier?

$$\tau_{cross} = \frac{L_A}{\langle v \rangle} = \frac{L_A}{\sqrt{\frac{k_B T}{2\pi m}}}$$

Stat Mech $\langle v \rangle$ from $\langle p \rangle$

$$\left(\langle p \rangle = \frac{\int_0^\infty p e^{-\frac{p^2}{2mk_BT}} dp}{\int_0^\infty e^{-\frac{p^2}{2mk_BT}} dp} \right)$$

$$= L_A \cdot \sqrt{\frac{2\pi m}{k_B T}}$$

✓ wider L_A longer

heavier particle longer ✓

higher T shorter ✓

$$= \sqrt{\frac{m k_B T}{2\pi}} \quad E_{kin} = \frac{p^2}{2m}$$

1D

2) How many particles can move?

rate of crossing

$$R_{cross} = \frac{[N^\# \text{ of particles at activated}]}{[\tau_{cross}]}$$

Similarly a particle spent $\begin{cases} \tau_A & \text{in activated} \\ \tau_W & \text{in well} \end{cases}$

$$N^\# \text{ at activated} = \frac{N \cdot \frac{\tau_A}{\tau_W + \tau_A + \tau_{cross}}}{\tau_{cross}} \approx \frac{N \cdot \frac{\tau_A}{\tau_W}}{\tau_{cross}}$$

Rate of crossing

$$R_{cross} = N \cdot \left(\frac{\tau_A/\tau_w}{\tau_{cross}}\right)$$ like 1st order reaction

Frequency

$$\Gamma' \text{ (jump freq)} = \boxed{\frac{\tau_A}{\tau_w}} \cdot \frac{1}{\tau_{cross}}$$

From Boltzmann dist

$$\tau_{cross} \Rightarrow \sqrt{\frac{2\pi m}{k_B T}} \cdot L_A$$

Length of activated / well states
Probability $\Big\} \Rightarrow \tau$

(trick: $\int \text{prob}\, dx \Rightarrow \tau$)

$$\frac{\tau_A}{\tau_w} = \frac{L_A}{L_w} e^{\left(-\frac{E_a}{k_B T}\right)}$$

Finally $$\Gamma' = \sqrt{\frac{k_B T}{2\pi m}} \cdot \frac{1}{L_w} e^{-\frac{E_a}{k_B T}}$$

length of well

No dependy on L_A any more

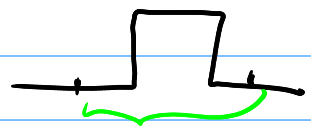
Arrhenius frequency

$$\nu = \sqrt{\frac{k_B T}{2\pi m}} \cdot \frac{1}{L_w} \Rightarrow \Gamma' = \nu e^{-\frac{E_a}{kT}}$$

To get D:

$$D = \frac{\Gamma \langle r^2 \rangle}{2}$$ (1D)

$\langle r^2 \rangle$?

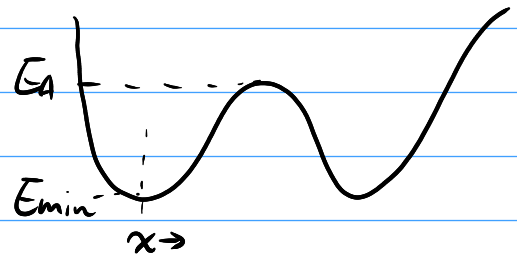


$r \approx L_{well} + L_A$

$$= \underbrace{\sqrt{\frac{k_B T}{8\pi m}} \frac{1}{L_w} \cdot (L_w + L_A)^2}_{D_0} \cdot \exp\left(-\frac{E_a}{k_B T}\right)$$

Very simplified, but works

More complex landscape?



Parabolic well

$$E(x) = E_{min} + \frac{\beta}{2}(x - x_{min})^2$$

$$\frac{\beta}{2}\left(\frac{L_w}{2}\right)^2 = E_a$$

$\Rightarrow$ modify Γ (or ν)

$$\Gamma' = \frac{1}{2\pi}\sqrt{\frac{\beta}{m}}\, e^{\left(-\frac{E_a}{k_B T}\right)}$$

$\Downarrow$

ν parabolic well

What are potentially missing?

1) One particle picture $\longrightarrow$ landscape can be "manybody"

2) Static landscape $\longrightarrow$ $E(\vec{x})$ depends on local env

3) Many particle system $\longrightarrow$ Same E_a diff configurations
   $\hookrightarrow$ Entropy effect

4) Real lattice $\longrightarrow$ symmetry factors to Γ

5) Correlated jump? $\longrightarrow$ f-factor

General solution for Γ' (KOM eq 7.25)

$$\Gamma' = \nu e^{-\frac{G^a}{k_B T}} = \nu e^{-\frac{H^a - TS^a}{k_B T}}$$

$$= \nu e^{\frac{S^a}{k_B}} e^{-\frac{H^a}{k_B T}}$$

$\nu = \frac{\omega_J}{2\pi}$ $\quad$ ω_J = angular frequency

Note: contribution from entropy

$$S^a = k_B \left[2\ln \frac{\omega_J}{\omega_J^A} + \sum_{l=1}^{3N^A} \ln\left(\frac{\omega}{\omega_i^A}\right) \right]$$

3D frequencies

$\Uparrow$

varies little in T

$\Longrightarrow$ Arrhenius equation $D = D_0 \exp\left(-\frac{H_a}{k_B T}\right)$

Activation enthalpy!

Fundation for theoretical calculations